

MT5 Strategy Analyzer

A professional Python application for analyzing **MetaTrader 5 (MT5)** backtest results, generating advanced strategy analytics, portfolio comparisons, and Excel reports.

Overview

MT5 Strategy Analyzer is a modular Python application designed to analyze MetaTrader 5 backtest reports and provide detailed statistical insights into trading strategies.

The application supports:

- Original strategy analysis
- L Pattern strategies (L, LL, LLL, LLLL, LLLLL)
- Drawdown (DD) based strategies
- Session-wise analysis
- Portfolio analysis
- Capital and Margin utilization
- Excel report generation
- Strategy optimization

The project is designed to become a reusable framework for analyzing any MT5 strategy.

Features

Original Strategy

- Trade Details
 - Monthly Analysis
 - Yearly Analysis
 - Equity Curve
 - Net Profit
 - MDD
 - Win Rate
 - Profit Factor
 - Recovery Factor
-

L Pattern Strategies

Supports:

- Original
- L
- LL
- LLL
- LLLL
- LLLLL

Features:

- Non-overlapping pattern detection
 - Independent strategy generation
 - Monthly/Yearly reports
 - Session reports
 - Trade Details
-

DD Strategies

Supports:

- DD10
- DD15
- DD20
- DD25
- DD30

Rolling Peak methodology.

Cycle detection includes:

- Trigger Date
 - Recovery Date
 - Recovery Days
 - Peak Margin
 - Capital Used
 - %ROE
-

Session Analysis

Supports:

- Asia
- Europe
- US Overlap
- Late US

User configurable.

Portfolio Analysis

Compare:

- Original
- All L Patterns
- All DD Strategies
- Custom portfolios

Metrics:

- Net Profit
 - MDD
 - Peak Margin
 - Capital Used
 - ROE
-

Optimization

Automatically identifies:

- Best Pattern
 - Best Session
 - Best Month
 - Best DD Strategy
 - Best Return/MDD
 - Best ROE
-

Excel Reports

The application generates formatted Excel workbooks.

Typical workbook includes:

- Dashboard
 - Summary
 - Monthly Analysis
 - Yearly Analysis
 - Session Analysis
 - Pattern Ranking
 - Trade Details
 - DD Cycles
 - Portfolio Comparison
-

Project Structure

```
MT5-Strategy-Analyzer
|
├── mt5_analyzer/
|   ├── core/
|   ├── domain/
|   ├── application/
|   ├── infrastructure/
|   ├── reports/
|   ├── gui/
|   └── utils/
|
├── tests/
├── samples/
├── output/
└── docs/
```

Requirements

- Python 3.12+
 - pandas
 - openpyxl
 - numpy
 - matplotlib
 - PySide6 (future GUI)
-

Installation

Clone repository

```
git clone https://github.com/<username>/MT5-Strategy-Analyzer.git
```

Create virtual environment

```
python -m venv .venv
```

Windows

```
.venv\Scripts\activate
```

Linux / macOS

```
source .venv/bin/activate
```

Install dependencies

```
pip install -e .
```

Usage

Analyze MT5 report

```
mt5-analyzer  
  --input "samples/MT5_Backtest.xlsx"  
  --output "./output"
```

Configuration

Application settings are stored in:

config/config.json

Configurable parameters include:

- Initial Equity
 - Margin per Trade
 - Lot Size
 - Time Zone
 - Sessions
 - Pattern Selection
 - DD Thresholds
-

Roadmap

v0.1

- Project Skeleton
 - MT5 Loader
 - Validator
 - Original Strategy
 - Statistics Engine
-

v0.2

- L Pattern Engine
-

v0.3

- Excel Reports
-

v0.4

- Session Engine
-

v0.5

- DD Engine

v0.6

- Portfolio Engine
-

v0.7

- Optimizer
-

v1.0

- Desktop GUI
 - Dashboard
 - Charts
 - PDF Reports
-

Development

Code quality tools:

- Black
- Ruff
- MyPy
- pytest

Every module includes:

- Type hints
 - Unit tests
 - Documentation
-

Contributing

Contributions are welcome.

Please:

1. Fork the repository
2. Create a feature branch

3. Commit your changes
4. Submit a Pull Request

License

MIT License

Author

MT5 Strategy Analyzer

Built using Python with a modular architecture for quantitative strategy research and MT5 backtest analysis.